The open source ecosystem needs regular funding, say community leaders

According to well-known members of the community, the open source ecosystem will soon require a regular stream of taxpayer financing to fix glaring resource deficiencies. A healthy software foundation will one day fall under the larger 'government purpose', with the public sector actively participating in stewardship, much like how maintaining electrical grids came into its scope a few centuries ago.

According to Amanda Brock, CEO of OpenUK, and Eric Brewer, VP of infrastructure at Google, who talked to IT Pro at State of Open Con 2023, long-standing financing issues in the open source community have widened the gap between heavily used but unmaintained packages that can contain vulnerable code and huge, well-maintained projects like Kubernetes.

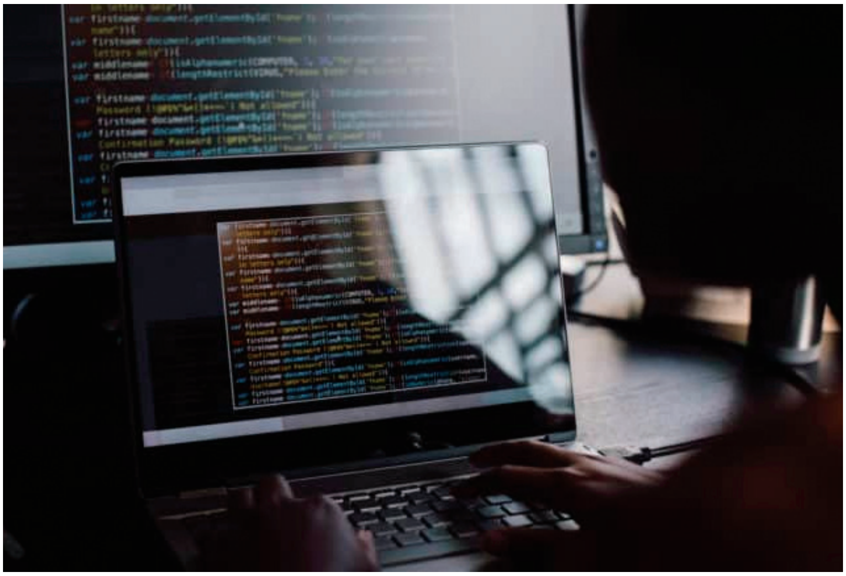
For instance, the Log4Shell exploit in 2021 addressed an undisclosed vulnerability in the widely used Log4j Java logging framework maintained by the Apache Software Foundation (ASF). Many people claimed the project should have been better supported since extra funding and code reviewers might have made a difference.

However, there are divides and arguments within the open source community regarding what the ideal finance and maintenance model might look like in the future, particularly to prevent future security horror stories.

For instance, Rebecca Rumbul, CEO of the Rust Foundation, told attendees at State of Open Con 2023 that governments shouldn't provide the majority or all of the funding for project maintenance. She said that more non-profit foundations, like her own, should be founded and funded in order to act as stewards for initiatives within the ecosystem, even though the public sector and businesses should both play some part.

VVenC and VVdeC H.266 video encoder and decoder now run on x86 and Arm

Open source H.266/VCC video encoder and decoder VVenC and VVdeC are both optimised for SIMD (single instruction, multiple data) instructions on x86 (SSE42/SIMDe and AVX2) and Arm, while the decoder is compatible with Windows, Linux, macOS, and Android. In 2020, the H.266 video compression standard, also known as VCC (versatile video coding), was approved with the promise of a 50 per cent data reduction over the previous H.265/HEVC standard while maintaining the same visual quality. Since that announcement, there have been no new developments, but the Realtek RTD1319D processor, which was unveiled last September and supports both 4K H.266 and AV1 video decoding, and the advancements made on the VVenC and VVdeC H.266 open source software encoder/decoder, which were discussed at FOSDEM 2023, may be changing that.



The Fraunhofer HHI group has been working on VVdeC and VVenC since the specifications were finalised in 2022. Both are based on VTM reference software for VCC, are written in C++ with a pure C interface, implement vectorization without the use of an assembler, and are provided under a BSD 3-Clause Clear licence that grants no patent rights. The source code for both is accessible on GitHub. VVdeC is fully Main10 profile compliant, supports more than 30 threads, runs on Windows, Linux (x86, Arm, RISC-V…), macOS (x86 and Arm), and Android. Since the first release, memory usage has been decreased by three times, and the developers are still making incremental advancements.

The VVenC open source H.266 encoder has five settings -- faster, fast, medium, slow, and slower -- each of which offers a balance between quality and encoding speed. It is designed for offline use and VoD (Video-on-Demand) operations. It is now possible to incorporate VVenC and VVdeC into FFmpeg via third-party patches, which enables inclusion into mpv, VLC, and ExoPlayer.

Imply Polaris wins 'Best Open Source Cloud Solution' award

Imply, the business established by the original developers of Apache Druid, announced that Imply Polaris has won the 'Best Open Source Cloud Solution' honour at The Cloud Awards, a global competition for cloud computing. Polaris offers an easy developer experience for creating real-time analytics apps, as a cloud database solution for Apache Druid.